

1. An equation is shown.

$$\frac{15}{7} + h = \frac{37}{7}$$

Complete the statement.

There is more than one correct way to complete the statement. See the assessment guide for additional correct response.

The value of h can be determined to be

- ☐ $\frac{15}{7}$
- ☒ $\frac{22}{7}$
- ☐ $\frac{37}{7}$
- ☐ $\frac{52}{7}$

because

- ☒ $\frac{15}{7}$
- ☐ $\frac{22}{7}$
- ☐ $\frac{37}{7}$
- ☐ $\frac{52}{7}$

is

- ☐ $\frac{15}{7}$
- ☒ $\frac{22}{7}$
- ☐ $\frac{37}{7}$
- ☐ $\frac{52}{7}$

- ☒ less than
- ☐ more than

- ☐ $\frac{15}{7}$
- ☐ $\frac{22}{7}$
- ☒ $\frac{37}{7}$
- ☐ $\frac{52}{7}$

2. Select all of the phrases that represent the algebraic expression shown.

$$y - 14$$

- ☒ fourteen less than y
- ☐ y less than fourteen
- ☒ the sum of y and -14
- ☐ y subtracted from -14
- ☒ the difference between y and 14

3. A candle weighs 300 grams (g). Each hour the candle is lit, it burns 4 g of its wax. Let h be the number of hours the candle burns.

Write an expression that represents the weight of the candle.

Expression	$300 - 4h$
-------------------	------------

Solve each equation.

4. $-6x = 108$

$$x = -18$$

5. $147 = b - (-12)$

$$b = 135$$

6. $w + 28 = 17$

$$w = -11$$

7. $-72 = \frac{y}{-2}$

$$y = 144$$

8. The Self-Transcendence 3,100 mile race is the world's longest certified footrace. Write an equation that can be used to determine how many miles a person has to run each day, x , if they complete the race in 50 days.

Equation	$50x = 3100$
-----------------	--------------

Solve each equation.

9. $k - 74 = -98$

10. $\frac{h}{25} = -9$

$k = -24$

$h = -225$

11. Buddy is planning a trip from Tampa to Jacksonville. He finds that traveling by car will take 204 minutes. He finds that traveling by train will take 112 minutes more than traveling by car.

- a. Write an equation that can be used to determine the time it takes to travel by train, x .

Equation	$x - 112 = 204$
-----------------	-----------------

- b. Use your equation to find the time it takes to travel by train. Show your work.

316 minutes

12. As the world's most expensive spice, saffron provides a beautiful golden color and pungent, aromatic flavor to foods. It has been used for centuries in Persian, Arab, European, and Indian cuisine. A package of saffron costs \$4 per gram. A restaurant buys a saffron package that weighs 75 grams. Which equation can be used to determine the cost of the package, c .

A. $c - 75 = 4$

B. $c + 4 = 75$

C. $\frac{c}{75} = 4$

D. $4c = 75$

13. Write an expression to represent the phrase “ n pounds of sugar equally separated into 8 cups”.

Expression	$\frac{n}{8}$
-------------------	---------------

14. The Marjorie Harris Carr Cross Florida Greenway is a hiking trail that traverses Citrus, Levy, Marion, and Putnam counties. The Greenway is 110 miles long. The Florida National Scenic Trail overlaps with the Greenway for 36 miles.

Write an equation that can be used to determine the number of miles, x , the Greenway does not overlap with the Florida National Scenic Trail?

Equation	$x + 36 = 110$
-----------------	----------------

15. A small submarine stops every 2.1 miles it descends below sea level. A bar model that represents the submarine’s distance in relation to sea level is shown.

-2.1	-2.1	-2.1	-2.1	-2.1
h				

Complete the statements.

The value of h in the bar model is

- ☐ -2.1.
- ☐ 5.
- ☒ -10.5.

The bar model represents the equation

- ☐ $-2.1h = -5.$
- ☒ $\frac{h}{5} = -2.1.$
- ☐ $h = -2.1.$